TARGET: NEET-UG PHYSICS & BOARD MASTERY

Electromagnetic Waves: Core Theory & Assessment

Complete Study Module: Displacement Current, Maxwell's Equations, Plane Waves, and Full Spectral Analysis

1. Overview and High-Yield Learning Objectives

In classical electromagnetism, Ampère's circuital law and Faraday's law of induction appeared to lack a symmetrical bridge: while changing magnetic flux generates an electric field, whether a changing electric field generates a magnetic field remained unanswered until James Clerk Maxwell identified the missing element. Maxwell introduced the concept of **displacement current**, resolving an inconsistency in Ampère's circuital law for non-steady currents.

This foundational insight unified electricity, magnetism, and optics, proving that light is a propagating transverse electromagnetic wave. For **NEET-UG**, questions consistently target:

- Calculation of displacement current i_d and its equivalence to conduction current i_c in capacitive circuits.
- Magnetic field calculation inside and outside parallel-plate capacitor geometries during charging/discharging.
- Vector relationship between electric field $\mathbf{E}$, magnetic field $\mathbf{B}$, and wave vector $\mathbf{k}$ ($\mathbf{E} \times \mathbf{B}$ points along wave propagation).
- Wave equations, amplitudes, speed relation $c = E_0/B_0 = 1/\sqrt{\mu_0 \varepsilon_0}$, and average energy densities.
- Order of frequencies and wavelengths in the electromagnetic spectrum, along with their production methods, detection mechanisms, and practical applications.

2. Inconsistency of Ampère's Law and Displacement Current

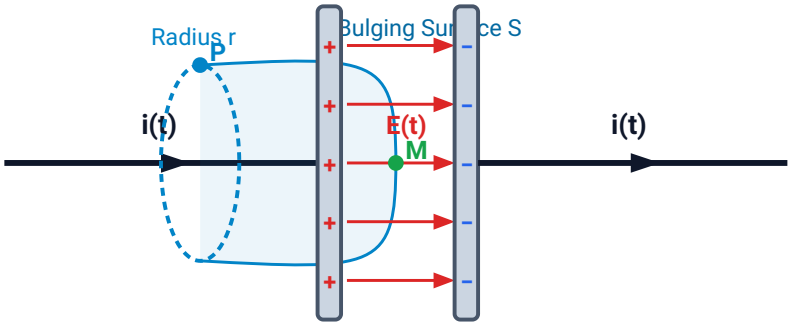
2.1 The Inconsistency in a Charging Capacitor

Consider a parallel-plate capacitor C connected to a time-varying source driving a charging current $i(t)$. Let us apply Ampère's circuital law, $\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = \mu_0 I_{\text{enclosed}}$, around a circular perimeter of radius r surrounding the lead wire at a point P outside the capacitor plates.

- **Surface 1 (Flat planar disk):** The lead wire pierces this surface. The conduction current passing through the surface is $i_c = i(t)$. Hence, $\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = 2\pi r B = \mu_0 i(t)$, giving a non-zero magnetic field.
- **Surface 2 (Pot-shaped or tiffin-box surface):** A bulging open surface whose rim is the exact same loop, but whose closed bottom passes entirely between the capacitor plates. No conducting wire penetrates this surface; hence, conduction current $i_c = 0$. This leads to $\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = 0$, implying $B = 0$ at point P.

This represents a direct contradiction: the value of the magnetic field at a single point P cannot depend on the choice of an arbitrary mathematical surface bounded by the same closed contour. Ampère's circuital law was incomplete because it omitted the effect of the time-varying electric field between the plates.

BOSONANCE



Continuity of Current:

- In wires: Conduction current
  ic = i(t), id = 0
- In gap: Displacement current
  ic = 0, id = $\varepsilon_0(d\Phi E/dt)$ = i(t)
- **Total current is continuous**

Tutorial Diagram 1: Resolution of Ampère's inconsistency using the displacement current concept across open surfaces bounded by the same contour.

2.2 Formulation of Displacement Current

If the parallel plates have plate area A and store charge $Q(t)$, the electric field magnitude inside the plates (ignoring fringing effects) is:

$$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$$

The electric flux Φ_E passing through a surface encompassing the entire cross-sectional plate area A between the plates is:

$$\Phi_E = E \cdot A = \left(\frac{Q}{\varepsilon_0 A}\right) A = \frac{Q}{\varepsilon_0}$$

Differentiating both sides with respect to time t:

$$\frac{\mathrm{d}\Phi_E}{\mathrm{d}t} = \frac{1}{\varepsilon_0}\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{i(t)}{\varepsilon_0} \implies \varepsilon_0 \frac{\mathrm{d}\Phi_E}{\mathrm{d}t} = i(t)$$

Maxwell designated this missing term as the **displacement current**, denoted by i_d:

MAXWELL'S DISPLACEMENT CURRENT

$$i_d = \varepsilon_0 \frac{\mathrm{d}\Phi_E}{\mathrm{d}t}$$

The total effective current through any open surface bounded by a closed loop is therefore $i = i_c + i_d$. This ensures complete continuity:

- Outside the capacitor plates: $i_c = i(t)$, while $i_d = 0$.
- Inside the dielectric gap: $i_c = 0$, while $i_d = i(t)$.
- The total current $i_{\text{total}} = i_c + i_d = i(t)$ is identical across all surfaces.

2.3 Ampère-Maxwell Law

Incorporating displacement current into Ampère's circuital law yields the generalized **Ampère-Maxwell Law**:

AMPÈRE-MAXWELL LAW

$$\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = \mu_0 \left(i_c + i_d\right) = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{\mathrm{d}\Phi_E}{\mathrm{d}t}$$

BOSONANCE

2.4 Magnetic Field Inside a Charging Circular Capacitor

Let a circular parallel-plate capacitor have radius R. A time-varying current $i(t)$ charges the plates uniformly. The displacement current density between the plates is uniform:

$$J_d = \frac{i_d}{\pi R^2} = \frac{i(t)}{\pi R^2}$$

By choosing a circular Amperian loop of radius r coaxial with the capacitor plates:

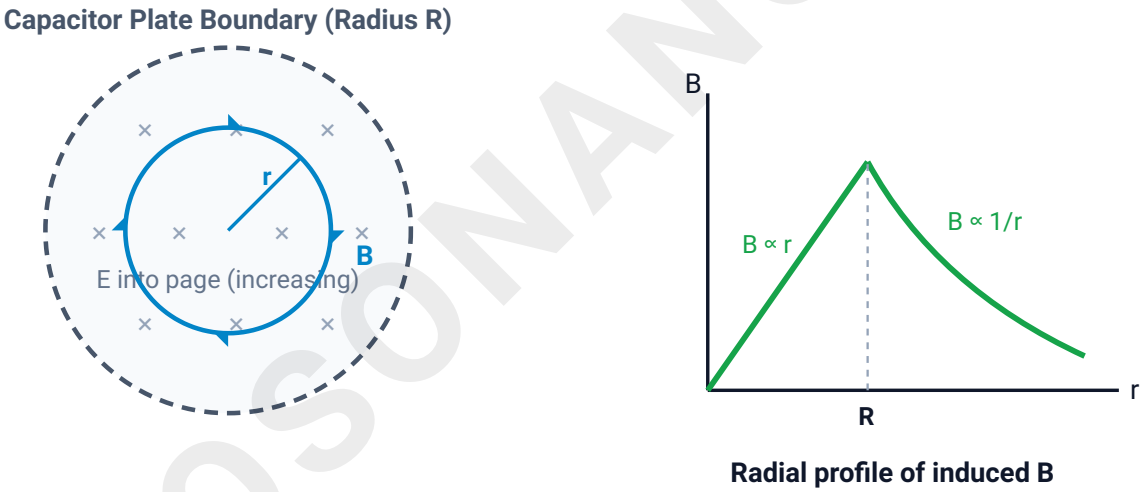
- **Inside the plates ($r \le R$):** The flux enclosed is $\Phi_E = E(\pi r^2)$. The enclosed displacement current is $i_{d,\text{enc}} = i_d \frac{\pi r^2}{\pi R^2} = i(t)\frac{r^2}{R^2}$. Applying Ampère-Maxwell's law:

  $$B(2\pi r) = \mu_0 i(t)\frac{r^2}{R^2} \implies B(r) = \frac{\mu_0 i(t) r}{2\pi R^2} \quad (r \le R)$$

  Thus, inside the plates, $B \propto r$. At the center axis ($r = 0$), $B = 0$.
- **Outside the plates ($r \ge R$):** The entire displacement current is enclosed: $i_{d,\text{enc}} = i(t)$.

  $$B(2\pi r) = \mu_0 i(t) \implies B(r) = \frac{\mu_0 i(t)}{2\pi r} \quad (r \ge R)$$

  Thus, outside the plates, $B \propto 1/r$, exactly matching the field around a long straight wire carrying conduction current $i(t)$.



Tutorial Diagram 2: Left: Induced concentric magnetic field lines between capacitor plates due to changing electric flux. Right: Variation of magnetic field magnitude with radial distance r.

BOSONANCE

3. Maxwell's Four Fundamental Equations in Vacuum

Maxwell compiled the fundamental laws of electrodynamics into four mathematical statements. Together with the Lorentz force law $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$, they fully govern classical electromagnetic phenomena:

Equation Name	Integral Form (in Free Space)	Physical Essence & Meaning
Gauss's Law for Electricity	$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$	Electric charges act as sources and sinks of electric fields. Total electric flux through any closed surface is proportional to enclosed net charge.
Gauss's Law for Magnetism	$\oint \mathbf{B} \cdot \mathrm{d}\mathbf{A} = 0$	Isolated magnetic monopoles do not exist. Magnetic field lines form continuous closed loops; net magnetic flux through any closed surface is zero.
Faraday's Law of Induction	$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}$	A time-varying magnetic flux produces an induced non-electrostatic electric field. The negative sign reflects Lenz's law.

Equation Name	Integral Form (in Free Space)	Physical Essence & Meaning
Ampère-Maxwell Law	$\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{\mathrm{d}\Phi_E}{\mathrm{d}t}$	Magnetic fields are generated both by moving electric charges (conduction current) and by time-varying electric fields (displacement current).

Symmetry Note for Competitive Examinations

Faraday's law shows that a time-varying magnetic field induces an electric field. Symmetrically, the Ampère-Maxwell law shows that a time-varying electric field induces a magnetic field. This mutual regeneration enables electromagnetic disturbances to propagate self-sustainingly across free space without a physical medium.

4. Production, Nature, and Properties of Electromagnetic Waves

4.1 Sources of Electromagnetic Radiation

- **Stationary Charges:** Produce only static electrostatic fields ($\mathbf{E} \neq 0, \mathbf{B} = 0$). No radiation.
- **Charges in Uniform Motion (Steady Currents):** Produce steady magnetic fields that do not change with time. No radiation.
- **Accelerating or Oscillating Charges:** An oscillating charge of frequency ν accelerates continuously. It generates time-varying electric and magnetic fields that feed into each other, radiating an electromagnetic wave into space of the exact same frequency ν.

4.2 Historical Experimental Milestones

- **Heinrich Hertz (1887):** Successfully produced and detected electromagnetic waves in the laboratory in the radio frequency range (wavelength $\sim$ a few metres), confirming Maxwell's theoretical predictions.
- **Jagdish Chandra Bose (Calcutta):** Generated and observed electromagnetic waves of very short wavelengths ranging from $25\ \mathrm{mm}$ down to $5\ \mathrm{mm}$ (millimetre waves / microwaves) in laboratory experiments.
- **Guglielmo Marconi:** Succeeded in transmitting electromagnetic waves over long distances (several kilometres), initiating modern wireless communication.

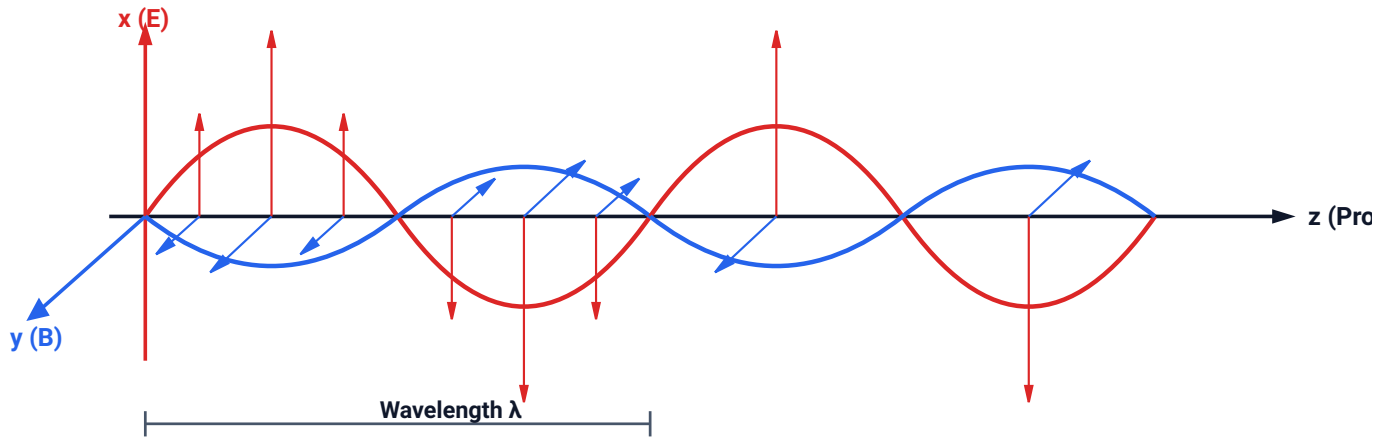
4.3 Transverse Nature and Vector Relations

Electromagnetic waves are strictly **transverse** in nature. The electric field vector $\mathbf{E}$ and magnetic field vector $\mathbf{B}$ oscillate mutually perpendicular to each other, and both are perpendicular to the direction of propagation vector $\mathbf{k}$.

DIRECTION OF WAVE PROPAGATION

$$\hat{\mathbf{v}} = \hat{\mathbf{k}} = \frac{\mathbf{E} \times \mathbf{B}}{|\mathbf{E} \times \mathbf{B}|}$$

BOSONANCE



Tutorial Diagram 3: A plane linearly polarized electromagnetic wave traveling in the $+z$ direction, with oscillating electric field along x and magnetic field along y.

4.4 Mathematical Representation of Plane Harmonic EM Waves

For a wave traveling in free space along the $+z$ direction:

$$E_x(z,t) = E_0 \sin(kz - \omega t) = E_0 \sin\left[2\pi\left(\frac{z}{\lambda} - \nu t\right)\right]$$

$$B_y(z,t) = B_0 \sin(kz - \omega t) = B_0 \sin\left[2\pi\left(\frac{z}{\lambda} - \nu t\right)\right]$$

where:

- $k = \frac{2\pi}{\lambda}$ is the propagation wave number (or propagation constant).
- $\omega = 2\pi\nu = \frac{2\pi}{T}$ is the angular frequency.
- $\nu\lambda = c$, where c is the speed of light in vacuum.

4.5 Speed, Energy Density, and Intensity Relations

KEY FORMULAS FOR NEET-UG

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3 \times 10^8 \,\text{m/s}, \qquad \frac{E_0}{B_0} = \frac{E_{\text{rms}}}{B_{\text{rms}}} = \frac{E}{B} = c$$

$$v_{\text{medium}} = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{\sqrt{\mu_r \varepsilon_r}} = \frac{c}{n} \quad (n = \text{refractive index})$$

$$\langle u_E \rangle = \frac{1}{4}\varepsilon_0 E_0^2, \qquad \langle u_B \rangle = \frac{B_0^2}{4\mu_0}$$

$$\text{Because } B_0 = \frac{E_0}{c} \text{ and } c^2 = \frac{1}{\mu_0\varepsilon_0} \implies \langle u_E \rangle = \langle u_B \rangle$$

$$\langle u_{\text{total}} \rangle = \langle u_E \rangle + \langle u_B \rangle = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0} = \varepsilon_0 E_{\text{rms}}^2$$

$$I = \langle u_{\text{total}} \rangle c = \frac{1}{2}\varepsilon_0 c E_0^2 = \varepsilon_0 c E_{\text{rms}}^2 = \frac{E_0 B_0}{2\mu_0}$$

Equipartition of Energy in EM Waves

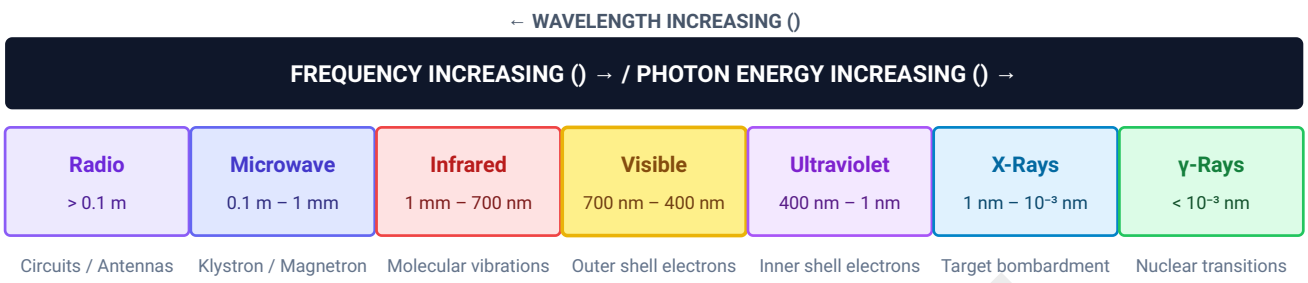
Although the magnitude of B_0 is numerically much smaller than E_0 (since $B_0 = E_0/c \approx E_0/(3 \times 10^8)$), the electric and magnetic fields carry **exactly equal shares** of the total average energy:

BOSONANCE

$$\langle u_E \rangle = \langle u_B \rangle = \frac{1}{2}\langle u_{\text{total}} \rangle.$$

5. The Comprehensive Electromagnetic Spectrum

The electromagnetic spectrum represents the orderly classification of electromagnetic waves according to their frequency or wavelength. There are no sharp, rigid boundaries between adjacent regions; classification is broadly based on how waves are produced and detected.



Tutorial Diagram 4: The electromagnetic spectrum arranged in order of decreasing wavelength and increasing photon energy / frequency.

5.1 Detailed Spectral Bands (NCERT Master Reference)

Type	Wavelength Range	Primary Production Method	Standard Detection	Key High-Yield Applications & Notes
Radio Waves	$> 0.1\,\text{m}$	Accelerated motion of electrons in conducting wires / aerials	Receiver aerials / circuits	• AM band: $530\,\text{kHz}$ to $1710\,\text{kHz}$ • Short wave bands: up to $54\,\text{MHz}$ • TV transmissions: $54\,\text{MHz}$ to $890\,\text{MHz}$ • FM band: $88\,\text{MHz}$ to $108\,\text{MHz}$ • Cellular telephony: Ultra High Frequency (UHF) band
Microwaves	$0.1\,\text{m}$ to $1\,\text{mm}$	Special vacuum tubes (klystrons, magnetrons, Gunn diodes)	Point contact diodes	• Aircraft navigation radar and automobile speed guns • Microwave ovens: frequency is tuned to match the resonant frequency of water molecules to heat food efficiently
Infrared (IR)	$1\,\text{mm}$ to $700\,\text{nm}$	Thermal vibrations of atoms and molecules in hot bodies	Thermopiles, bolometers, infrared photographic film	• Often called *heat waves* because water, CO_2, and NH_3 absorb them strongly • Physical therapy lamps, remote controls for TV/audio systems • Greenhouse effect: maintains Earth's warmth • Satellite imaging for military reconnaissance and crop growth
Visible Light	$700\,\text{nm}$ to $400\,\text{nm}$ (4×10^{14} to $7 \times 10^{14}\,\text{Hz}$)	Atomic transitions of valence electrons from higher to lower levels	Human retina, photocells, photographic film	• Narrow band detected by human eye (center of sensitivity matches solar peak) • Snakes detect infrared; many insects detect into ultraviolet
Ultraviolet (UV)	$400\,\text{nm}$ to $1\,\text{nm}$ (down to $0.6\,\text{nm}$)	Inner shell electron transitions; very hot bodies, electric welding arcs, Sun	Photocells, photographic film	• Absorbed by atmospheric ozone layer (40–$50\,\text{km}$ altitude) • Absorbed by ordinary glass (welding masks use special glass;

BOSONANCE

Type	Wavelength Range	Primary Production Method	Standard Detection	Key High-Yield Applications & Notes
				no tanning through glass windows) • Induces melanin production (sunburn/tanning) • Focused beams used in LASIK eye surgery; germicidal UV lamps purify drinking water
X-Rays	$1\,\text{nm}$ to $10^{-3}\,\text{nm}$ ($10^{-8}\,\text{m}$ to $10^{-13}\,\text{m}$)	Bombardment of heavy metal target with high-energy electrons; inner shell transitions	Photographic film, Geiger tubes, ionization chambers	• Medical diagnosis for bone fractures and dental imaging • Radiation therapy for certain forms of cancer • High ionizing potential damages living tissue; excessive exposure must be avoided
Gamma Rays (γ)	$< 10^{-3}\,\text{nm}$ ($10^{-10}\,\text{m}$ to $< 10^{-14}\,\text{m}$)	Nuclear reactions and radioactive decay of unstable atomic nuclei	Photographic film, Geiger-Müller counters, ionization chambers	• Highest frequency, highest penetrating power, highest photon energy • Used in radiotherapy to selectively destroy malignant cancer cells

6. Step-by-Step Worked Representative Examples

Worked Example 1: Parallel-Plate Capacitor Charging

Displacement Current & Induced B

A parallel plate capacitor consists of two circular plates each of radius $R = 12\,\text{cm}$ separated by a distance $d = 5.0\,\text{mm}$. It is charged by an external constant current of $0.15\,\text{A}$.

1. Calculate the capacitance C and the rate of change of potential difference $\frac{\text{d}V}{\text{d}t}$ across the plates.
2. Determine the displacement current i_d passing through the inter-plate region.
3. Find the magnitude of the induced magnetic field between the plates at a distance $r = 6.0\,\text{cm}$ from the central axis.

Solution:

Part 1: The plate area is $A = \pi R^2 = \pi(0.12\,\text{m})^2 \approx 4.524 \times 10^{-2}\,\text{m}^2$.
Using $\varepsilon_0 = 8.85 \times 10^{-12}\,\text{C}^2/(\text{N}\cdot\text{m}^2)$ and $d = 5.0 \times 10^{-3}\,\text{m}$:

$$C = \frac{\varepsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 4.524 \times 10^{-2}}{5.0 \times 10^{-3}} \approx 8.01 \times 10^{-11}\,\text{F} = 80.1\,\text{pF}$$

Since $q = CV$, differentiating with respect to time gives $i = C\frac{\text{d}V}{\text{d}t}$. Thus:

$$\frac{\text{d}V}{\text{d}t} = \frac{i}{C} = \frac{0.15\,\text{A}}{8.01 \times 10^{-11}\,\text{F}} \approx 1.87 \times 10^{9}\,\text{V/s}$$

Part 2: Inside the dielectric gap, the displacement current is equal to the conduction current in the connecting wires:

$$i_d = \varepsilon_0 \frac{\text{d}\Phi_E}{\text{d}t} = \varepsilon_0 A \frac{\text{d}E}{\text{d}t} = \frac{\varepsilon_0 A}{d}\frac{\text{d}V}{\text{d}t} = C\frac{\text{d}V}{\text{d}t} = i = 0.15\,\text{A}$$

Part 3: At $r = 6.0\,\text{cm} = 0.06\,\text{m}$, which is inside the plates ($r < R = 0.12\,\text{m}$):

$$B = \frac{\mu_0 i_d r}{2\pi R^2} = \left(\frac{\mu_0}{2\pi}\right)\frac{i_d r}{R^2} = \left(2 \times 10^{-7}\right)\frac{0.15 \times 0.06}{(0.12)^2} = 2 \times 10^{-7} \times \frac{0.009}{0.0144} = 1.25 \times 10^{-7}\,\text{T}$$

BOSONANCE

Worked Example 2: Plane EM Wave Equation and Vector Analysis

Wave Properties

The magnetic field component of an electromagnetic plane wave in vacuum is given by:

$$B_y = (3 \times 10^{-7}\,\text{T}) \sin\left[(1.5 \times 10^{3}\,\text{rad/m})x + (4.5 \times 10^{11}\,\text{rad/s})t\right]$$

1. Determine the direction of propagation, the wavelength λ, and the frequency ν.
2. Write the corresponding vector expression for the electric field $\mathbf{E}$.

Solution:

Part 1: The argument of the sine function is $(kx + \omega t)$. A plus sign indicates that the wave propagates along the **negative x-direction ($-\hat{\mathbf{i}}$)**.

Wave number $k = 1.5 \times 10^{3}\,\text{rad/m} \implies \lambda = \frac{2\pi}{k} = \frac{2\pi}{1.5\times10^{3}} \approx 4.19 \times 10^{-3}\,\text{m} = 4.19\,\text{mm}.$

Angular frequency $\omega = 4.5 \times 10^{11}\,\text{rad/s}$

$\implies \nu = \frac{\omega}{2\pi} = \frac{4.5\times10^{11}}{2\pi} \approx 7.16 \times 10^{10}\,\text{Hz} = 71.6\,\text{GHz}.$

(Note: $v = \omega/k = (4.5 \times 10^{11})/(1.5 \times 10^{3}) = 3 \times 10^{8}\,\text{m/s} = c$).

Part 2: Amplitude of electric field:

$$E_0 = cB_0 = (3 \times 10^{8}\,\text{m/s}) \times (3 \times 10^{-7}\,\text{T}) = 90\,\text{V/m}$$

For propagation direction $\hat{\mathbf{k}} = -\hat{\mathbf{i}}$ and $\mathbf{B}$ oscillating along the y-axis ($\hat{\mathbf{B}} = \hat{\mathbf{j}}$), the unit vector of the electric field $\hat{\mathbf{E}}$ must satisfy:

$$\hat{\mathbf{E}} \times \hat{\mathbf{B}} = \hat{\mathbf{k}} \implies \hat{\mathbf{E}} \times \hat{\mathbf{j}} = -\hat{\mathbf{i}}$$

Since $\hat{\mathbf{k}} \times \hat{\mathbf{j}} = -\hat{\mathbf{i}}$, the electric field must oscillate along the **$+z$-direction ($\hat{\mathbf{k}}$)**.

Thus, the complete vector equation is:

$$\mathbf{E}(x,t) = \hat{\mathbf{k}}\,(90\,\text{V/m}) \sin\left[(1.5 \times 10^{3}\,\text{rad/m})x + (4.5 \times 10^{11}\,\text{rad/s})t\right]$$

7. NEET-UG Pattern Practice Assessment

The following questions are prepared strictly in accordance with current **NEET-UG** testing patterns, encompassing single-option multiple choice questions, Statement-I and Statement-II evaluations, Assertion-Reasoning, and Matrix Matching.

BOSONANCE

Practice Question 1

NEET Single Correct MCQ

A parallel plate capacitor with circular plates of radius R is being charged by a steady current I. The magnitude of the magnetic field at a distance r from the central axis in the region between the plates is:

(A) Directly proportional to r for $r < R$ and inversely proportional to r for $r > R$

(B) Inversely proportional to r for all values of r

(C) Zero everywhere because no actual physical charge flows between the plates

(D) Constant and independent of r for $r < R$

Correct Answer: (A)

Worked Reasoning: Between the capacitor plates, conduction current is zero, but a time-varying electric flux gives rise to a uniform displacement current density $J_d = \frac{I}{\pi R^2}$. For an Amperian loop of radius $r \leq R$, enclosed displacement current is $I_{\text{enc}} = I\frac{r^2}{R^2}$. From $\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = B(2\pi r) = \mu_0 I \frac{r^2}{R^2} \implies B = \frac{\mu_0 I r}{2\pi R^2} \propto r$. For $r \geq R$, all displacement current is enclosed, yielding $B = \frac{\mu_0 I}{2\pi r} \propto \frac{1}{r}$.

Practice Question 2

NEET Numerical MCQ

The amplitude of the electric field in a plane electromagnetic wave traveling in vacuum is $E_0 = 120\,\mathrm{N/C}$. If the wave has a frequency of $50.0\,\mathrm{MHz}$, the amplitude of the magnetic field B_0 and the propagation constant k are respectively:

(A) $4.0 \times 10^{-7}\,\mathrm{T}$ and $1.05\,\mathrm{rad/m}$

(B) $3.6 \times 10^{10}\,\mathrm{T}$ and $0.52\,\mathrm{rad/m}$

(C) $4.0 \times 10^{-7}\,\mathrm{T}$ and $2.09\,\mathrm{rad/m}$

(D) $1.2 \times 10^{-6}\,\mathrm{T}$ and $1.05\,\mathrm{rad/m}$

Correct Answer: (A)

Worked Reasoning:
1. The magnetic field amplitude is given by:

$$B_0 = \frac{E_0}{c} = \frac{120}{3 \times 10^8} = 4.0 \times 10^{-7}\,\mathrm{T}$$

2. The propagation wave number is:

$$k = \frac{\omega}{c} = \frac{2\pi\nu}{c} = \frac{2\pi \times (50 \times 10^6)}{3 \times 10^8} = \frac{100\pi \times 10^6}{300 \times 10^6} = \frac{\pi}{3} \approx 1.047 \approx 1.05\,\mathrm{rad/m}$$

BOSONANCE

Practice Question 3

NEET Statement-Based Question

Given below are two statements:

Statement I: An electric charge moving with a constant velocity produces a steady magnetic field but cannot emit electromagnetic waves.

Statement II: In a plane electromagnetic wave propagating in free space, the average energy density of the electric field is much greater than that of the magnetic field because $E_0 \gg B_0$.

In light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Correct Answer: (C)

Worked Reasoning:

• **Statement I is correct:** An accelerating charge is necessary to radiate electromagnetic waves. Stationary charges produce only electrostatic fields, while charges moving at constant velocity produce steady magnetic fields that do not vary with time and do not radiate.

• **Statement II is incorrect:** The average energy densities are exactly equal: $\langle u_E \rangle = \frac{1}{4}\varepsilon_0 E_0^2$ and $\langle u_B \rangle = \frac{B_0^2}{4\mu_0}$. Substituting $E_0 = cB_0$ and $c = 1/\sqrt{\mu_0 \varepsilon_0}$ proves $\langle u_E \rangle = \langle u_B \rangle$. The electric and magnetic fields share energy equally.

BOSONANCE

Practice Question 4

NEET Vector Analysis MCQ

An electromagnetic wave travels in vacuum along the $+y$-direction. At a certain instant, the electric field vector is directed along the $+z$-axis. What is the direction of the magnetic field vector $\mathbf{B}$ at this instant?

(A) Along the $+x$-direction

(B) Along the $-x$-direction

(C) Along the $+y$-direction

(D) Along the $-z$-direction

Correct Answer: (B)

Worked Reasoning: The wave propagation direction is along the vector product $\mathbf{E} \times \mathbf{B}$. Here, the propagation vector is $\hat{\mathbf{k}}_{\text{prop}} = \hat{\mathbf{j}}$, and the electric field direction is $\hat{\mathbf{E}} = \hat{\mathbf{k}}$. We must find unit vector $\hat{\mathbf{B}}$ such that:

$$\hat{\mathbf{k}} \times \hat{\mathbf{B}} = \hat{\mathbf{j}}$$

Testing unit vectors along coordinate axes:

$$\hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \quad \text{and} \quad \hat{\mathbf{k}} \times (-\hat{\mathbf{i}}) = -\hat{\mathbf{j}}$$

Wait: using cyclic order $\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}$, $\hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}$, and $\hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$. Hence $\hat{\mathbf{k}} \times \hat{\mathbf{i}} = +\hat{\mathbf{j}}$, so $\hat{\mathbf{B}}$ must be along $+\hat{\mathbf{i}}$? Wait! Let's re-verify: $\mathbf{E} \times \mathbf{B}$: If $\mathbf{E}$ is along $+z$ ($\hat{\mathbf{k}}$) and $\mathbf{B}$ is along $-x$ ($-\hat{\mathbf{i}}$), then $\hat{\mathbf{k}} \times (-\hat{\mathbf{i}}) = -(\hat{\mathbf{k}} \times \hat{\mathbf{i}}) = -(\hat{\mathbf{j}})$. Wait, let's re-evaluate: $\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}$ $\hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}$ $\hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$. So if $\mathbf{E}$ is along $+z$ ($\hat{\mathbf{k}}$) and propagation is along $+y$ ($\hat{\mathbf{j}}$), then $\hat{\mathbf{k}} \times \hat{\mathbf{B}} = \hat{\mathbf{j}}$. Since $\hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$, $\hat{\mathbf{B}}$ is along $+\hat{\mathbf{i}}$ (i.e. $+x$-direction)! Wait, let us check option (A): Along $+x$-direction. Let's verify: $\mathbf{E} \times \mathbf{B} = E\hat{\mathbf{k}} \times B\hat{\mathbf{i}} = EB(\hat{\mathbf{k}} \times \hat{\mathbf{i}}) = EB\hat{\mathbf{j}}$. Indeed, $EB\hat{\mathbf{j}}$ points in the $+y$ direction! Therefore, $\mathbf{B}$ is along the $+x$-direction.

BOSONANCE

Practice Question 5

NEET Assertion & Reason

Assertion (A): Welder's face masks are fitted with special glass goggles during electric arc welding.
Reason (R): Electric welding arcs emit large quantities of ultraviolet (UV) radiation which can damage the eyes, and ordinary or specially treated glass absorbs UV rays effectively.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Correct Answer: (A)

Worked Reasoning: Electric arcs operate at extremely high temperatures and emit high-intensity ultraviolet radiation. Prolonged exposure causes serious ocular tissue damage. Because glass strongly absorbs ultraviolet radiation, special glass face masks are mandatory personal protective equipment for welders. Both statements are true and Reason (R) directly explains Assertion (A).

BOSONANCE

Practice Question 6

NEET Spectral Applications MCQ

Match List-I (Electromagnetic wave type) with List-II (Characteristic production device / application):

List-I	List-II
(P) Microwaves	(1) Diagnostic tool to detect bone fractures and cancer treatment
(Q) Infrared waves	(2) LASIK precision eye surgery and water purification
(R) Ultraviolet rays	(3) Klystrons, magnetrons, and aircraft radar navigation
(S) X-rays	(4) Greenhouse effect and physical therapy lamps

(A) P-3, Q-4, R-2, S-1

(B) P-3, Q-2, R-4, S-1

(C) P-1, Q-4, R-2, S-3

(D) P-4, Q-3, R-1, S-2

Correct Answer: (A)

Worked Reasoning:

- Microwaves (P) are generated by klystrons/magnetrons and used in radar $\rightarrow$ (3).
- Infrared waves (Q) cause greenhouse warming and are used in therapeutic lamps $\rightarrow$ (4).
- Ultraviolet rays (R) have high focusability for LASIK and germicidal action $\rightarrow$ (2).
- X-rays (S) are used in diagnostic bone imaging and cancer radiotherapy $\rightarrow$ (1).

BOSONANCE

Practice Question 7

NEET Energy & Radiation Pressure

An electromagnetic wave of intensity $I = 1.327 \times 10^3\,\mathrm{W/m^2}$ is incident normally upon a completely absorbing flat surface of area $A = 1.5\,\mathrm{m^2}$. If the radiation is incident continuously for $t = 20\,\mathrm{minutes}$, the total momentum transferred to the surface is:

(A) $7.96 \times 10^{-3}\,\mathrm{kg \cdot m/s}$

(B) $3.98 \times 10^{-3}\,\mathrm{kg \cdot m/s}$

(C) $2.39 \times 10^{6}\,\mathrm{kg \cdot m/s}$

(D) $1.59 \times 10^{-2}\,\mathrm{kg \cdot m/s}$

Correct Answer: (A)

Worked Reasoning:

Total energy incident on the surface in time $t = 20\,\mathrm{min} = 1200\,\mathrm{s}$ is:

$$U = I \times A \times t = 1.327 \times 10^3 \times 1.5 \times 1200 = 2.3886 \times 10^6\,\mathrm{J}$$

For a completely absorbing surface, the linear momentum transferred is:

$$p = \frac{U}{c} = \frac{2.3886 \times 10^6\,\mathrm{J}}{3.0 \times 10^8\,\mathrm{m/s}} \approx 7.96 \times 10^{-3}\,\mathrm{kg \cdot m/s}$$

BOSONANCE

Practice Question 8

NEET Frequency & Medium Calculation

An electromagnetic wave of frequency $\nu = 2.0 \times 10^{10}\,\text{Hz}$ travels from vacuum into a non-magnetic medium having relative permittivity $\varepsilon_r = 4.0$ and relative permeability $\mu_r = 1.0$. Its velocity, wavelength, and frequency in the medium will be:

(A) $v = 1.5 \times 10^{8}\,\text{m/s}$, $\lambda = 0.75\,\text{cm}$, $\nu = 2.0 \times 10^{10}\,\text{Hz}$

(B) $v = 3.0 \times 10^{8}\,\text{m/s}$, $\lambda = 1.50\,\text{cm}$, $\nu = 2.0 \times 10^{10}\,\text{Hz}$

(C) $v = 1.5 \times 10^{8}\,\text{m/s}$, $\lambda = 1.50\,\text{cm}$, $\nu = 1.0 \times 10^{10}\,\text{Hz}$

(D) $v = 0.75 \times 10^{8}\,\text{m/s}$, $\lambda = 0.375\,\text{cm}$, $\nu = 2.0 \times 10^{10}\,\text{Hz}$

Correct Answer: (A)

Worked Reasoning:

1. Refractive index $n = \sqrt{\mu_r \varepsilon_r} = \sqrt{1.0 \times 4.0} = 2.0$.
2. Speed in the medium: $v = \frac{c}{n} = \frac{3 \times 10^8}{2} = 1.5 \times 10^{8}\,\text{m/s}$.
3. The frequency ν is a characteristic of the source and **remains unchanged**: $\nu = 2.0 \times 10^{10}\,\text{Hz}$.
4. Wavelength in the medium: $\lambda = \frac{v}{\nu} = \frac{1.5 \times 10^8}{2.0 \times 10^{10}} = 0.75 \times 10^{-2}\,\text{m} = 0.75\,\text{cm}$.

8. Common Student Pitfalls & High-Yield Revision Checklist

Frequent Conceptual Traps in Competitive Exams

1. **Confusing E/B with B/E:** Remember that $c = E_0/B_0$ has dimensions of velocity (m/s). Therefore, $E_0 = cB_0$, never $B_0 = cE_0$.
2. **Assuming Displacement Current Requires Conducting Medium:** Conduction current requires free mobile charge carriers (electrons or ions). Displacement current requires *only* a time-varying electric field and exists readily in absolute vacuum.
3. **Misidentifying Phase Difference:** In a plane vacuum electromagnetic wave, $\mathbf{E}$ and $\mathbf{B}$ reach their peaks, zero-crossings, and troughs at the exact same coordinates and instants in time. The phase difference between $\mathbf{E}$ and $\mathbf{B}$ is **zero** ($\Delta\phi = 0$), although their spatial vectors are oriented at 90° to each other.
4. **Wavelength Inversion in Medium:** Frequency is strictly invariant during refraction across media; velocity and wavelength both decrease by the factor $n = \sqrt{\mu_r \varepsilon_r}$.
5. **Ozone Layer Altitude:** The protective stratospheric ozone layer is located at an altitude of approximately 40 to $50\,\text{km}$.

Final Rapid Revision Checklist

- [ ] **Displacement Current:** $i_d = \varepsilon_0 \frac{\mathrm{d}\Phi_E}{\mathrm{d}t}$. Continuous with i_c in circuits.
- [ ] **Internal Magnetic Field of Capacitor:** $B(r) = \frac{\mu_0 i_d r}{2\pi R^2}$ for $r \le R$; $B(r) = \frac{\mu_0 i_d}{2\pi r}$ for $r \ge R$.
- [ ] **Wave Speed:** $c = 1/\sqrt{\mu_0 \varepsilon_0} = 3 \times 10^8\,\text{m/s}$; in medium $v = 1/\sqrt{\mu \varepsilon}$.
- [ ] **Energy Balance:** Electric energy density equals magnetic energy density ($\langle u_E \rangle = \langle u_B \rangle$). Total average energy density $\langle u \rangle = \frac{1}{2}\varepsilon_0 E_0^2$.

BOSONANCE

- [ ] **Propagation Vector:** $\hat{\mathbf{k}} \parallel \mathbf{E} \times \mathbf{B}$.
- [ ] **Spectrum Sequence (Decreasing λ):** Radio $\rightarrow$ Microwave $\rightarrow$ Infrared $\rightarrow$ Visible $\rightarrow$ UV $\rightarrow$ X-rays $\rightarrow$ γ-rays.

9. Concise Chapter Summary

Maxwell completed the mathematical framework of classical electromagnetism by discovering that a time-dependent electric field acts as a source of magnetic field through **displacement current** ($i_d = \varepsilon_0 \mathrm{d}\Phi_E/\mathrm{d}t$). This restored consistency to Ampère's law and demonstrated that coupled, time-varying electric and magnetic fields self-propagate as transverse electromagnetic waves with velocity $c = 1/\sqrt{\mu_0 \varepsilon_0}$.

Accelerated electric charges radiate electromagnetic waves whose frequency matches the oscillation frequency of the charge. The electromagnetic spectrum extends seamlessly from long radio waves ($\lambda > 0.1\,\mathrm{m}$) to high-energy gamma rays ($\lambda < 10^{-12}\,\mathrm{m}$). Mastery of wave vectors, energy equipartition, and production/detection mechanisms provides a complete foundation for full scoring in NEET-UG physics assessments.

BOSONANCE